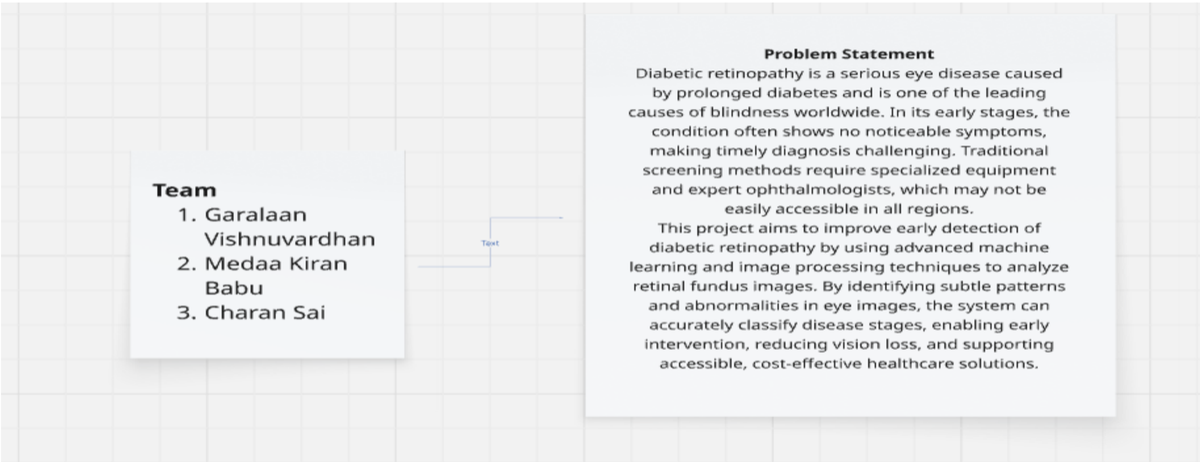
Ideation Phase
Brainstorm & Idea Prioritization Template

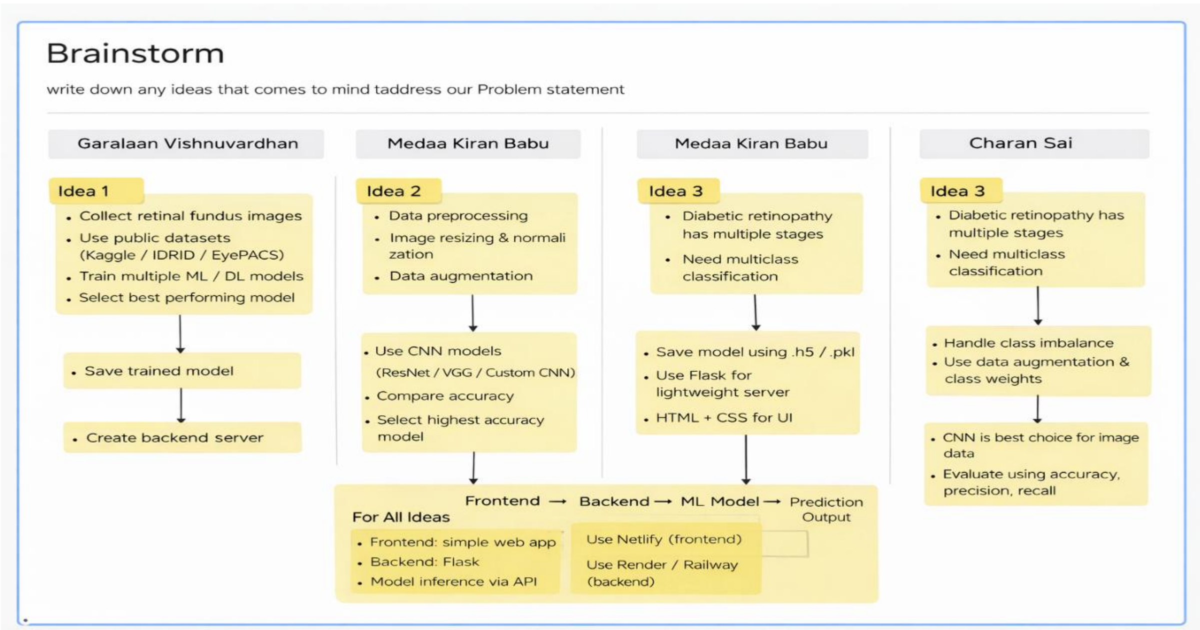
Date	14 feb 2026
Team ID	LTVIP2026TMIDS35912
Project Name	Deep Learning Fundus Image Analysis for Early Detection of Diabetic Retinopathy
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

Step-1: Team Gathering, Collaboration and Select the Problem Statement



Step-2: Brainstorm, Idea Listing and Group



Brainstorm – Group Idea Flow

1. Data Collection

- Collect retinal fundus images
- Use publicly available datasets (Kaggle, IDRiD, EyePACS)
- Ensure sufficient data for all disease stages

2. Data Preprocessing

- Image resizing and normalization
- Data augmentation to increase dataset size
- Improve model generalization

3. Problem Understanding

- Diabetic retinopathy has multiple stages
- Requires multiclass classification
- High-dimensional image data

4. Model Selection

- Train multiple machine learning / deep learning models
- Prefer CNN-based architectures (ResNet, VGG, Custom CNN)
- Compare models based on accuracy and performance
- Select the best-performing model

5. Handling Challenges

- Address class imbalance in dataset
- Use data augmentation and class weighting
- Reduce overfitting and noise

6. Model Training & Saving

- Train the final selected model
- Save the trained model using .pt format
- Prepare model for inference

7. Backend Development

- Build a lightweight backend server using django
- Load the trained model
- Create API for prediction

8. Frontend Development

- Design a simple and user-friendly web interface
- Use HTML and CSS for UI
- Allow image upload for prediction

9. System Integration

- Connect frontend with backend

- Perform model inference through API
- Display prediction results clearly

10. Deployment

- Deploy frontend as a web application
- Deploy backend on cloud platform
- Ensure global accessibility

11. Final Output Flow

Frontend → Backend → ML Model → Prediction Output

Step-3: Idea Prioritization